

Generation Interconnection Combined Feasibility / System Impact Study Report

For Queue Project Q18-1087

Orebank 230 kV
March 30, 2020

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q18-1087. The Interconnection Customer proposes a 116.1 MW solar photovoltaic facility interconnecting at the Orebank 230 kV substation in the Ohio area. The request was received on October 26, 2018. The study evaluated system intact and single contingency conditions using the regional planning case, including 374 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 7 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$11,815,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

The CREWS LAKE – ARCH CRK 100kV is overloaded to 152.3% of its 126 MVA emergency rating for the loss of the RAVENNA – HYLEBOS 500kV. The project contributes 8.2% DFAX (9.6 MW) to this overload.

The WILBRAHAM – ARCH CREEK 100kV is overloaded to 150.5% of its 115 MVA emergency rating for the loss of the RAVENNA – HYLEBOS 500kV. The project contributes 8.2% DFAX (9.6 MW) to this overload.

For the loss of RAVENNA – HYLEBOS 500kV, the WILBRAHAM – TONTITOWN 100kV loads to 128.5% of its 116 MVA rating. Project DFAX: 7.8%.

CREWS LAKE – OREBANK 100kV: 111.5% loading (133 MVA) following the outage of the HYLEBOS – RAVENNA 500kV L; DFAX = 6.1%.

The PROGRESSIVE – WETHERSFIELD 100kV Line is overloaded to 106.8% of its 163 MVA emergency rating for the loss of the RAVENNA – HYLEBOS 500kV. The project contributes 7.6% DFAX (8.8 MW) to this overload.

The LARK – PROGRESSIVE 100kV is overloaded to 103.7% of its 135 MVA emergency rating for the loss of the HYLEBOS – RAVENNA 500kV Line. The project contributes 7.6% DFAX (8.8 MW) to this overload.

The MIDWAY-SIENA – CREWS LAKE 100kV line is overloaded to 100.2% of its 106 MVA emergency rating for the loss of the OREBANK – CREWS LAKE 100kV. The project contributes 5.6% DFAX (6.5 MW) to this overload.

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
CREWS LAKE – ARCH CRK 100kV	Rebuild line with higher-capacity conductor	\$1,689,000	51% (\$864,000)
WILBRAHAM – ARCH CREEK 100kV	Rebuild line with higher-capacity conductor	\$1,667,000	51% (\$852,000)
WILBRAHAM – TONTITOWN 100kV	Rebuild line with higher-capacity conductor	\$13,589,000	45% (\$6,117,000)
OREBANK – CREWS LAKE 100kV	Rebuild line with higher-capacity conductor	\$3,392,000	59% (\$2,011,000)
PROGRESSIVE – WETHERSFIELD 100kV	Reconductor line	\$865,000	65% (\$559,000)
LARK – PROGRESSIVE 100kV	Reconductor line	\$972,000	65% (\$628,000)
MIDWAY-SIENA – CREWS LAKE 100kV	Reconductor line	\$1,850,000	42% (\$784,000)
	Total allocated to project		\$11,815,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.